DoRA: Weight-Decomposed Low-Rank Adaptation

CS 4/5782 Deep Learning

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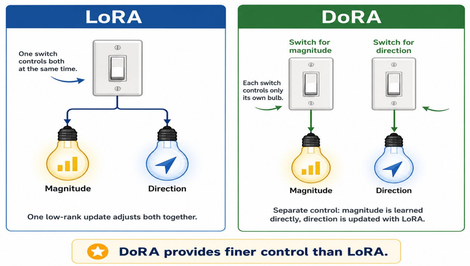
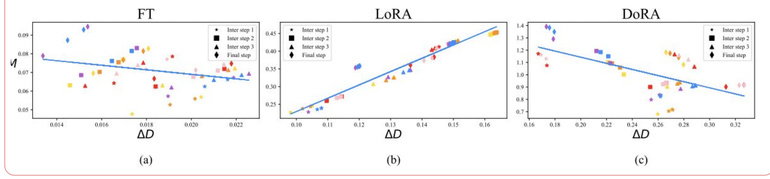
References

1. Introduction

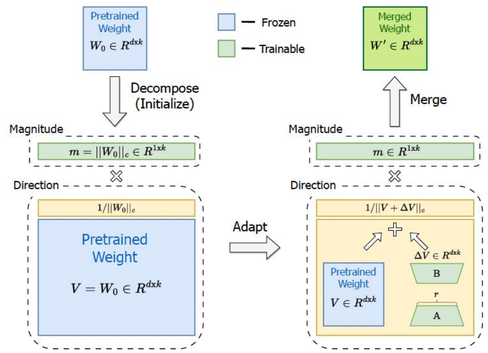
LoRA is efficient, but it doesn't learn like Full-finetuning





2. Methodology

DoRA decomposes the pre-trained weight matrix W0 into its magnitude and direction, applies low-rank adaptation to direction, and merges them to obtain the adapted weight We.



Example Dataset

- Multi-task commonsense reasoning datasets
- PIQA, BoolQ, SIQA, HellaSwag, WinoGrande, ARC, OBQA
- Each sample: question + multiple-choice answers
- Task: select the correct answer

3. Results

DoRA achieves improved accuracy compared to LoRA on eight commensense reasoning tasks

	Method	rank	Params(%)	BoolQ	PIQA	SIQA	HellaSwag	WinoGrande	Arc-e	Arc-c	OBQA	Avg
LLaMA-7B	LoRA	16	0.6700	61	76.5	69.5	71	77	77.5	57	68.5	69.8
	DoRA*	16	0.4274	66.5	74.5	77	72.5	78.5	75	60	74.5	**72.3**
	DoRA	32	0.8384	64.5	73.5	75	70	76	77.5	59	68	70.5
LLaMA-13B	LoRA	16	0.6700	68	80.5	82.5	71	80.5	82.5	67.5	79.5	76.4
	DoRA*	32	0.3467	67	80.5	80	77.5	80	82.5	70	79.5	77.1
	DoRA	32	0.6806	71	77.5	82.5	79.5	83	85.5	65.5	79	**77.9**
LLaMA2-7B	LoRA	16	0.6700	70	75.5	80.5	71.5	79.5	83	63	75	74.8
	DoRA*	32	0.4274	70	74	80	74.5	79	83.5	68.0	80.5	76.2
	DoRA	32	0.8384	68.5.	76.5	79.5	77.5	79	81	69.5	79	**76.3**
LLaMA3-8B	LoRA	16	0.6700	70.5	80.5	78	84.5	79	90	78	78	79.8
	DoRA*	32	0.4274	70	80.5	76.5	84.5	85.5	90	79	80	80.8
	DoRA	32	0.8384	70	81.5	78.5	83	84.5	92	81	83	**81.7**

The Mighty DoRA

4. What Else can DoRA do? → Diffusion

Base Model: Stable Diffusion XL (SDXL 1.0)

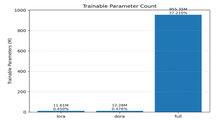
Method: apply DoRA to attention layers of SDXL UNet, rank = 8

Dataset: Traditional Chinese Ink-wash landscape paintings

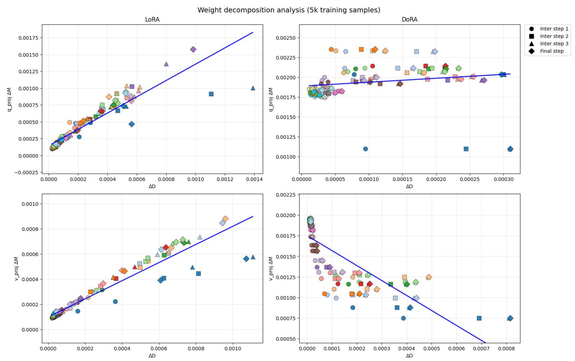
Pros:

1. Minimal compute

   < 0.5% of all trainable params!



2. Weight direction and magnitude decoupling



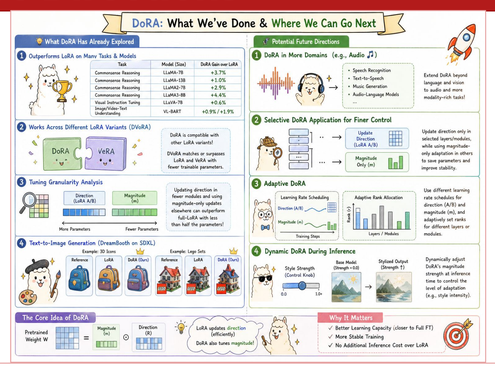
5. Extension: RLHF

Base Model: LLaMA-7B

Method: Compare LoRA and our custom DoRA under the same m

DPO-based RLHF setup on Anthropic HH-RLHF

Dataset: Anthropic HH-RLHF

	Method	rank r	Raw LogP Acc.	DPO Reward Acc.	Raw LogP Margin	DPO Reward Margin
0	LoRA	16	46.2	51.2	-5.9456	0.0332
1	LoRA	32	47.6	55.9	-5.4257	0.0852
2	DoRA	16	46.7	52.6	-5.5950	0.0683
3	DoRA	32	47.2	54.8	-5.3953	0.0883

6. Future Directions